37 A detector is placed near a radioactive isotope and records a count rate of 700 counts / min.

The half-life of the isotope is 8 min.

The average background count rate is 60 counts / min.

What is the count rate measured by the detector after 16 min?

A 100 counts / min

B 160 counts / min

C 175 counts / min

D 220 counts / min